

DESIGN AND DEVELOPMENT OF A ROBOTIC ARM WITH REASSEMBLING CAPABILITY

A thesis submitted in partial fulfilment of the requirements for the award of the
degree of

Bachelor of Technology

in

Electrical and Electronics Engineering

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APRIL 2022

BONAFIDE CERTIFICATE

This is to certify that the project titled
**DESIGN AND DEVELOPMENT OF A ROBOTIC ARM WITH
REASSEMBLING CAPABILITY**

is a bonafide record of the **EE402 Project Phase II** work done by

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in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Electrical and Electronics Engineering** of the **NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY**, during the year 2021-2022.

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ABSTRACT

In this present world, technology has become an invaluable part. We can witness the advancement and development in many scientific domains. Currently, the field of robotics is transforming and evolving at a very fast pace. Contemporary robots are being used in place of humans to do jobs which are dangerous, boring or require high accuracy. A wide range of robots have been developed and being implemented in various fields.

This project is aimed to design a robotic arm which can be completely assembled and reassembled. The arm is being modelled in such a way that the dimensions (size) of the arm can be altered according to the user convenience. This variation is achieved by using removable modules (which have in-built components) instead of static metallic parts. Each module consists of a stepper motor, motor driver, copper windings and carbon brushes to transmit current. Stepper motor converts a train of pulses to a precisely defined increment in shaft's rotation. Motor driver is used to enable stepper motor the way it works. In-order to control the position and step angle Arduino board is implemented. Coordination between two modules is done by serial interfacing of Arduino board in the joints.

The robotic arm is given with a gripper to grab objects of varied sizes which is controlled with Augmented Reality technology. OpenCV is a cross-platform library using which we can develop real-time computer vision applications. It mainly focuses on image processing, video capture and analysis including features like face detection and object detection. Here Open CV is used to detect palm landmarks, which are used to vectorize and calculate the joints if the gripper does grab and releasing action. From Open CV command is being send to Arduino NANO which is used for the coordinated operation of the system. In addition to these a servo motor is connected in-order to facilitate position control.

This robotic arm can be operated in Remote and Local modes. Since the dimensions of this model can be varied, it could be used efficiently.

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CHAPTER 1

INTRODUCTION

1.1 General Introduction

A Robotic arm is a Type of mechanical arm, usually programable, with similar functions to a human arm. It can be used in many fields of day to day works to industrial automation. Each moduls are connected by joints allowing either rotaional or translational motion.

Advantages of Robots

- **More Efficiency:** Robots can do certain tasks faster and more efficiently than humans because they are designed and built to perform them more precisely.
- **Working in Dangerous Environments:** Robotic automation is the ideal solution in extreme working conditions in order to eliminate the need for people to expose themselves to unnecessary risks.
- **Better Quality:** Due to their greater precision, industrial robots can be used to produce higher quality products resulting in less time.
- **Mapping Out Disaster Zones:** Disaster zones are scary and dangerous places. Unmanned flying robots can be sent into these no-go zones and reveal the situation.

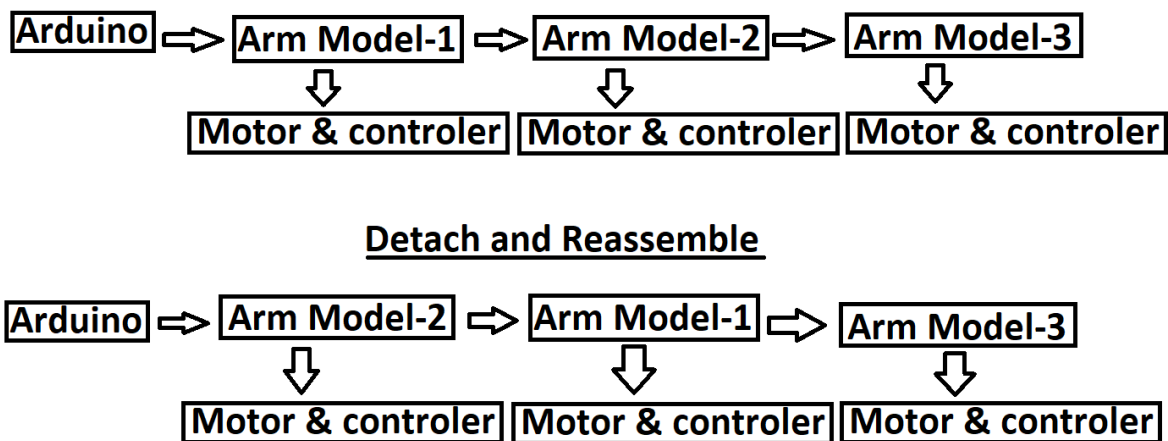
Disadvantages of Robots

- **Skilled man power:** Highly skilled man power is needed to implement and maintain robots. This will increase the initial and operational costs of the companies which use robots in their operations.
- **High cost:** Operational costs are higher due to power costs and grease costs.
- **Chances of malfunction:** If the robots are not well maintained, it will malfunction, this may be disastrous during manufacturing or production processes.

Challenges in Robotics

- i. Manufacturing procedures in robotics haven't seen many changes since ages. When it comes to hardware, a large number of industrial and commercial robots are still made rigid and unstable.
- ii. Robots can carry out a single task efficiently. However, a single robot cannot perform multiple functions with the same efficiency or efficacy.
- iii. Robots that are deployed in industrial settings are usually static and carry out a single repetitive task.

1.2 Architecture



1.3 Objective

The main objective of the project is to design and develop a robotic arm which can be assembled and reassembled with an ease. This cannot be achieved by conventional robotic assembly which is done with rigid metallic parts. Rather, individual modules are developed with certain specifications and are assembled to a complete structure.

1.4 Motivation

Robotics is a complex field of study and construction of robots alone is another difficult task. Very broadly, the problem of robotic assembly consists in designing a computer-controlled robotic system.

The two major drawbacks of conventional assembly system are

- reprogramming
- lack of flexibility.

Once in a completely assembled robot it is hard to make any alternations.

This made us to think of a robotic arm, whose dimensions can be altered according to the users wish. This can be achieved, if the robotic arm is made out of combination of modules rather than solid metallic parts.

1.5 Outline of the thesis

Chapter 1 gives the introduce us to robotic arm, its process and the reason to make it re-assembled and how we are making it.

Chapter 2 shows the Literatures which we took as reference to help us with the idea and the process involved in it.

Chapter 3 shows the output of the simulation of the prototype to make the Outer Frame of the Model.

Chapter 4 explains the Concept and application of Stepper motor using Arduino to the robotic arm.

Chapter 5 explains the Atmega32 and how to use it to make a serial communication and how to assign the order of operation.

Chapter 6 deals with the conclusion of this thesis.

CHAPTER-2

LITERATURE REVIEW

2.1 Review of Coupling Mechanism Designs for Modular Reconfigurable Robots

Wael Saab, Peter Racioppo and Pinhas Ben-Tzvi* Robotics and Mechatronics Laboratory, Department of Mechanical Engineering, Virginia Tech, Blacksburg, VA 24061, USA.

With the increasing demands for versatile robotic platforms capable of performing a variety of tasks in diverse and uncertain environments, the needs for adaptable robotic structures have been on the rise. In the last two decades, researchers in the field of modular reconfigurable robotics have developed novel coupling mechanisms intended to establish rigid and robust connections, while enhancing system autonomy and reconfigurability.

This paper is related to robotic coupling mechanism designs, with the aim of outlining current progress and identifying key challenges and opportunities that lay ahead.

2.1.1 Pick and Place Robotic Arm Implementation Using Arduino

Ashly Baby, Chinnu Augustine, Chinnu Thampi, Maria George, Abhilash A P, Philip C Jose (IOSR-JEEE) 2017.

The soft catching gripper used in the arm will not apply any extra pressure on the objects.

In this journal, robotic arm is designed using Arduino to pick and place the objects via user commands. It will pick and place an object from source to destination safely. The robot is controlled using android based smart phones through Bluetooth. Based on the commands given by the user the robot moves accordingly. At the receiver end there are four motors interfaced with the micro controller. Two for the vehicle movement and the remaining two are for arm and gripper movement. Blue control application is used for the controlling of robot.

2.2 Speed Control of The Direct Current Servomotor And The Stepper Motor With Arduino Uno Platform

E Darie, R Pécsi and M Culcea, the 7th Conference of the Sustainable Solutions for Energy and Environment IOP Conf. Series: Earth and Environmental Science 664 (2021) 012055.

Servomotors are special direct current or alternating current electric machines which have an adjustable rotational speed in a wide range, in both directions. Stepper motors are special synchronous machines that have a different construction than the classic one and are used in precision drive systems, with open loop.

This paper experiences the speed control of a direct current servomotor, type SM-S2309S, with the help of the Arduino UNO platform, which is the simplest solution for the development of electronic applications. The study experiences also speed and step control of a stepper motor, type 28BYJ-48, with the Arduino UNO platform.

2.3 Augmented Reality-Assisted Robot Programming System for Industrial Applications

Mechanical Engineering Department, Faculty of Engineering, National University of Singapore, 9 Engineering Drive 1, Singapore 117576, Singapore

This paper discusses an Augmented Reality (AR) assisted robot programming system (ARRPS) that provides faster and more intuitive robot programming than conventional techniques. ARRPS is designed to allow users with little robot programming knowledge to program tasks for a serial robot.

With an AR user interface and a handheld pointer for interaction, users are free to move around the work cell to define 3D points and paths for the real robot to follow. Sensor data and algorithms are used for robot motion planning, collision detection and plan validation.

2.4 A Review of Augmented Reality in Robotic-Assisted Surgery

Long Qian, Student Member, IEEE, Jie Ying Wu, Student Member, IEEE, Simon DiMaio, Nassir Navab, Member, IEEE, and Peter Kazanides, Member, IEEE.

Augmented reality (AR) and robotic-assisted surgery (RAS) are both rapidly evolving technologies in recent years. RAS systems aim to improve surgical precision and dexterity, as well as access to minimally-invasive procedures, while AR provides an advanced interface to enhance user perception.

This paper discusses the hardware components, application paradigms and clinical relevance of the literature. The concept of AR-integrated RAS has been shown to be feasible for various procedures. Encouraging preliminary results include reduced sight diversion and improved situation awareness. Special techniques like activation-on-demand, are taken into consideration to address visual clutter of the AR interface and ensure that the system is fail-safe.

2.5 Layout of the Construction

This segment gives an overall idea of step-by-step procedure carried out in our project starting from stimulating the prototype to assembling the joints and exhibiting the movement of robotic arm with 3 degrees of freedom (DoF).

Step 1: Designing the prototype in Solid works software.

Step 2: Designing the outer frame with suitable material in 3D printer.

Step 3: Incorporating winding, slip rings and brushes in order to facilitate current flow in each module.

Step 4: Installing stepper motor and motor drive of required specifications.

Step 5: Giving speed control and position control to the motor using Arduino UNO board.

Step 6: Serial interfacing of Arduino board present in each module.

Step 7: Assembling the joints into a complete module.

Step 8: Adding a gripper tip to the arm and controlling it with AR technology.

CHAPTER 3

FABRICATION OF THE PROTOTYPE

3.1 Unity Software

Unity is a cross-platform game engine developed by Unity Technologies, which is primarily used to develop video games and simulations for computers, consoles and mobile devices. Unity is the tool used by a large number of game developers to create and power their creations. This software is powerful, extremely easy to use, and free (Students' version). Simulating prototypes help us in getting a pictorial representation of our Ideology. Moreover, we can remodel the design and upgrade each time as per our wish.

In-order to have a better idea about the hardware design and to analyse the operating conditions virtually a prototype is simulated in Unity software. Modules are built using building blocks with required dimensions and are joined in such a way that the modules can facilitate rotating movement. In each module a set of five slip ring like structures are incorporated, which serve the purpose of current conduction. Brush (carbon) like structures is attached to the slip rings which facilitate in tapping the current.

Individual modules are then attached at certain axis in order to get a complete structure of the arm. To have considerable rotation motion certain angles are given at each joint. In this project our aim is to have 3 degrees of freedom in the robotic arm. After giving all the specifications the model is set to simulate the output and observed. We observe that the modules of the arm are free to rotate at desired angles without any obstacles. Also, we observed that the detachment and reassembling of modules from one another do not affect the inbuilt structures of slip rings and brushes.

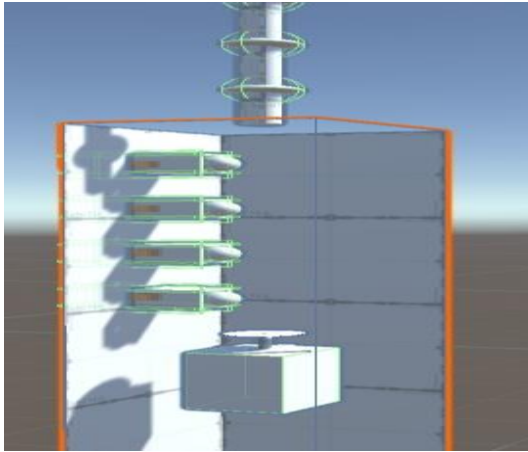


Fig. 3.1.1 Slip rings when adjacent modules are not in contact.

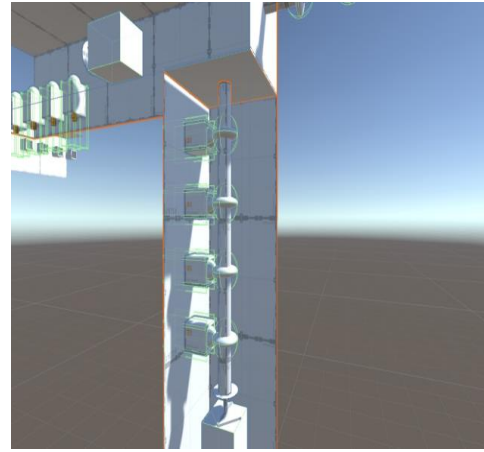


Fig. 3.1.2 slip ring positions when adjacent modules are in contact.

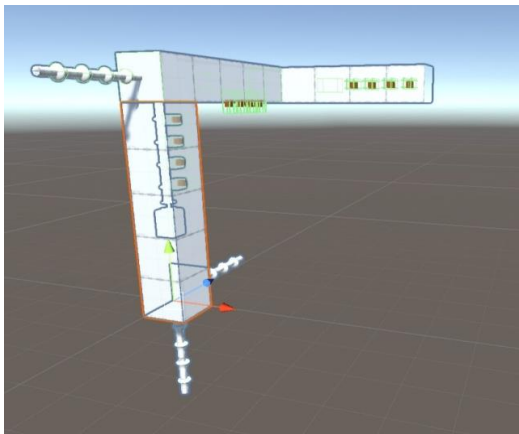


Fig. 3.1.3 Degree of rotation (0,0,0)

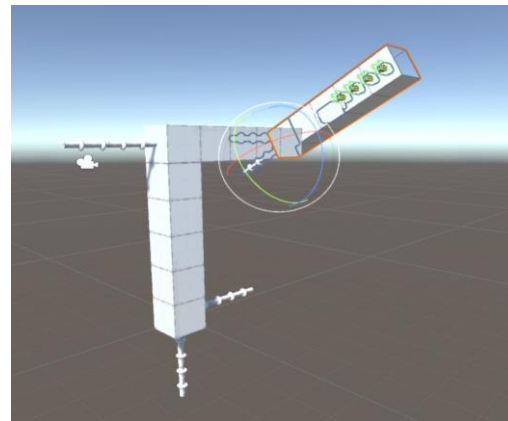


Fig. 3.1.4 Degree of rotation (0,0,29.63)

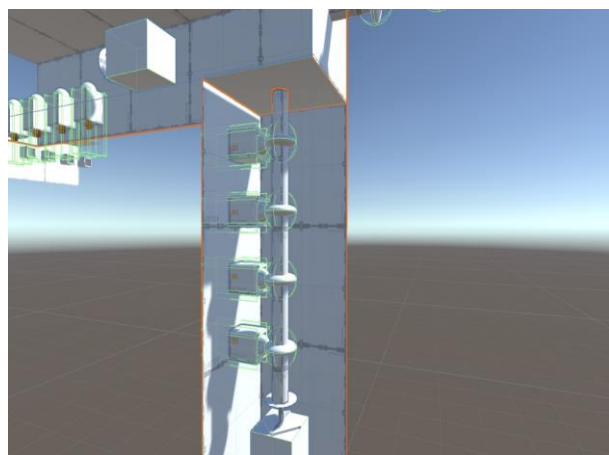


Fig.3.1.5 Degree of rotation (0,16.88,29.63)

3.2 Designing of outer frame

A virtual model is designed in unity software to have a clear idea while proceeding further.

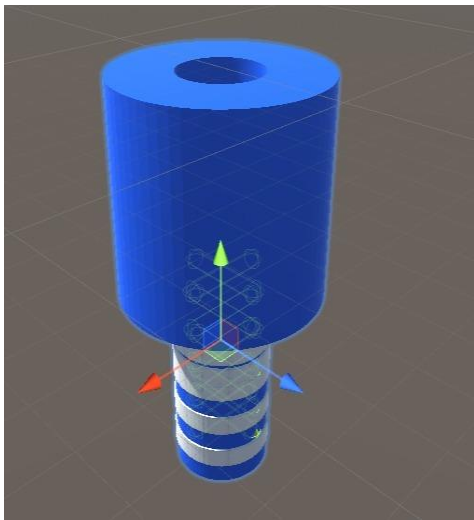


Fig. 3.2.1 Part 1

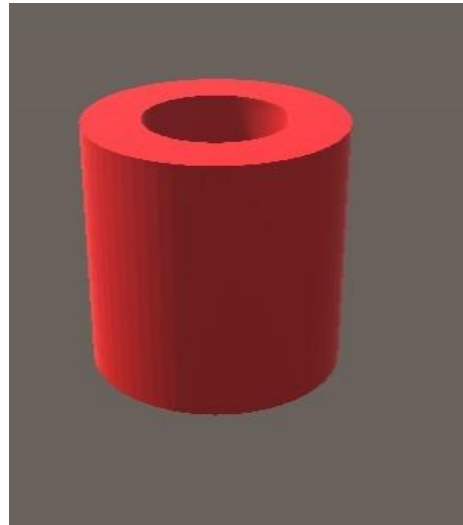


Fig.3.2.2 Part 2

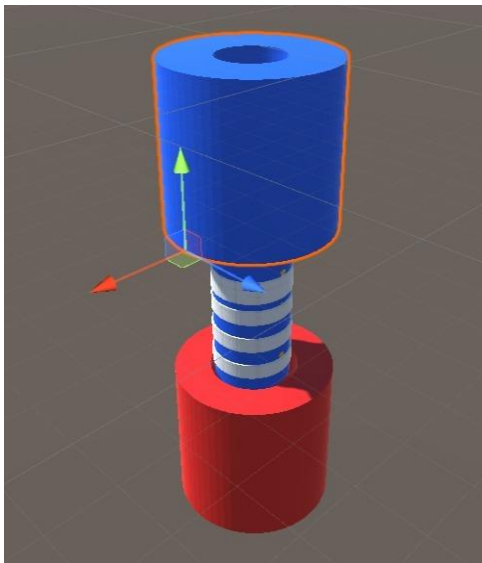


Fig.3.2.3 Housing part 1 over part 2

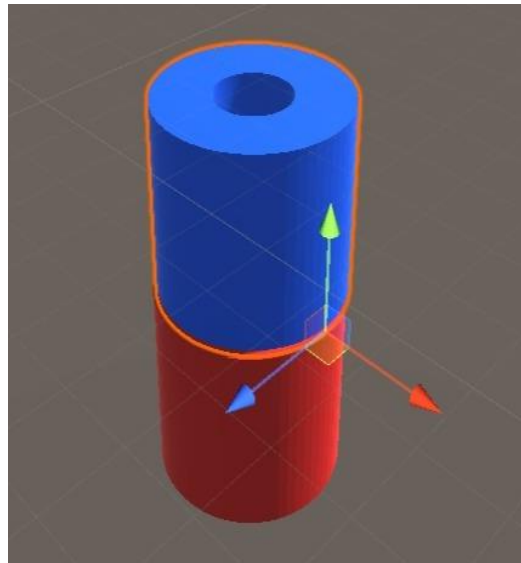


Fig.3.2.4 A complete module

It was planned to fabricate the outer frame in 3D printing, since it is more reliable and have high precision in the model. But by keeping the cost ineffectiveness we decided to model the prototype with wood. The prototype is designed in wood and its movement is observed.

3.3 Outer Frame of The Robotic Arm Made Out Of Wood



Fig. 3.3.1 Part 1



Fig. 3.3.2 Part 2



Fig. 3.3.3 Housing Part 1 Over Part 2



Fig 3.3.4 Complete joint

On trying to have rotating motion over the joints of module, we found that the rotation of the module is not smooth due to inertia of wood. It was also observed that friction is more in between the joints and this opposes the free rotation of the module. In order to have perfection

in movement to frame should have smooth surface. Weight of the frame should be optimised else the motor will face difficulty while rotating.

3.4 Incorporating winding and brushes

Modules attached to a complete arm should be coordinated and electrically coupled. Insulated copper wire of considerable thickness is wound over the surface of the joint in order to have conduction of current in the rotating joint. Carbon brushes are inserted at required positions to tap current from copper winding. The choice for carbon brushes is due to its ready availability in market and its flexibility. Moreover, these brushes are self-lubricating type, one of the best electrical conductors which increase the conductivity.

After the insertion of brushes to the copper winding, the conduction of current is checked using multimeter. It is observed that a considerable amount of current flow is there in the winding. The significance of this copper winding is to act as a conduction path in between the rotating joints. Moreover, this winding facilitates the current conduction in between the modules. To give mechanical support to the brushes an extension structure – brush holder is designed.

Instead of copper wire, a strip of copper can also be used which reduces the possibility of shorting in between the windings. This ideology is in consideration and it can be implemented in the model while proceeding further developments.

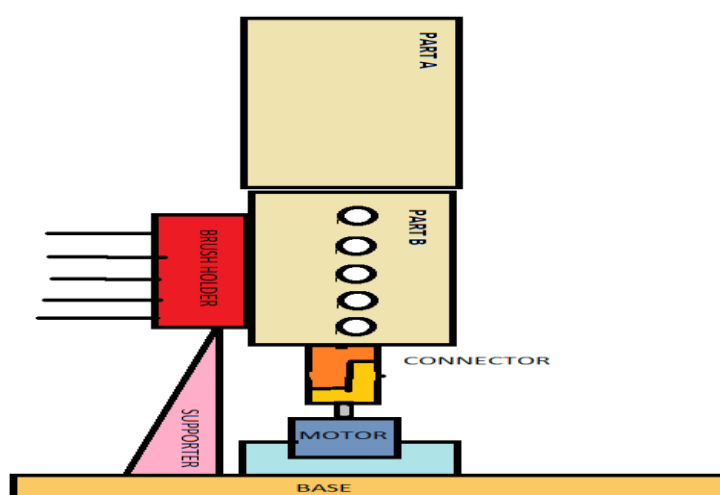


Fig 3.4.1 Diagram of base module



Fig 3.4.2 PVC Pipe Model (Front view)



Fig 3.4.3 PVC Pipe Model (Side view)

3.5 Designing and Simulation in Solid Works Software

SolidWorks is a solid modelling computer-aided design and computer-aided engineering application. This software is used to develop mechatronics systems from beginning to end. At the initial stage, the software is used for planning, visual ideation, modelling, feasibility assessment, prototyping, and project management. The software is then used for design and building of mechanical, electrical, and software elements. Finally, the software can be used for management, including device management, analytics, data automation, and cloud services.

The Solid works software solutions are used by mechanical, electrical, and electronics engineers to form a connected design. The suite of programs is aimed at keeping all engineers in communication and able to respond to design needs or changes.

Solid works Dfm Tolerance Analysis allows the user to foresee manufacturing as early on as the design stage whilst ensuring hole alignment and reducing material waste to a minimum. Interference Detection is suitable for both static and dynamic assemblies allowing any defects to be highlighted prior to production. The core and cavity tools allow for auto generation of parting lines, draft angles and other such features to enable design users to create accurate mould products. Moulds can be created directly from core and cavity geometry within a model and set to industry standards.

The software's assembly modelling feature allows designers and engineers to run collision detection from within the design space. Mould tools are able to auto generate draft lines as well as core and cavity tools, draft angles can be quickly and easily checked to ensure parts will work within moulds.

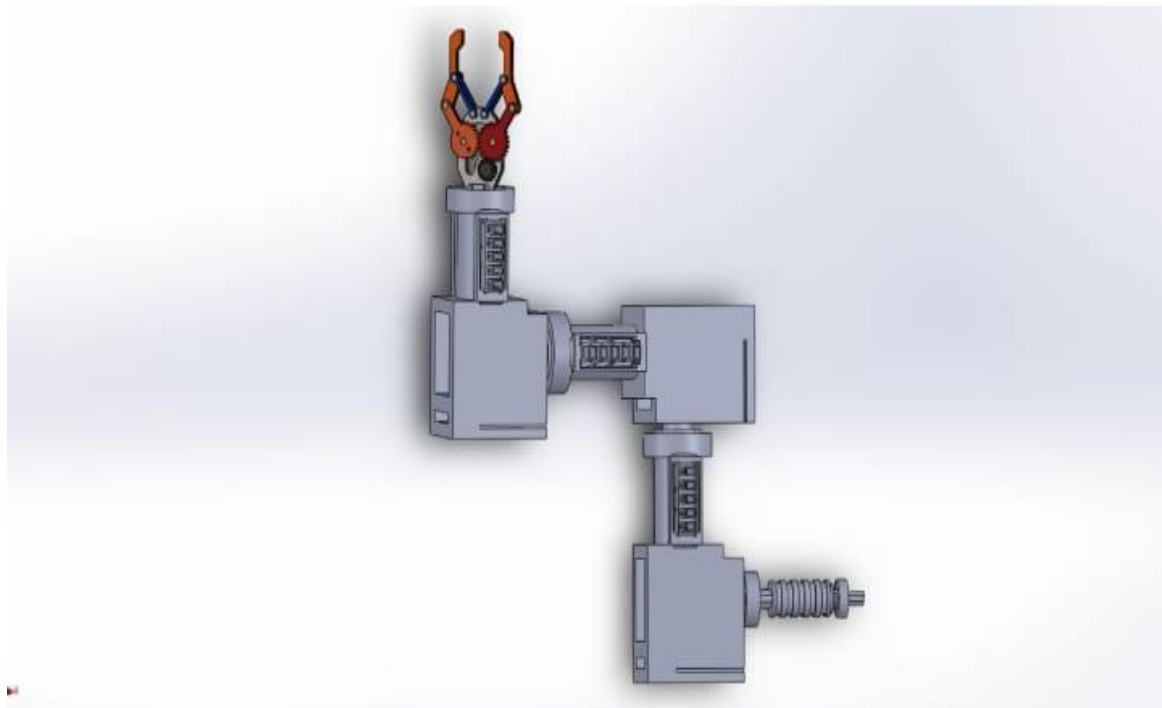


Fig 3.5.1 Robotic arm simulated in solidworks software



Fig 3.5.2 Axis of rotation of robotic arm

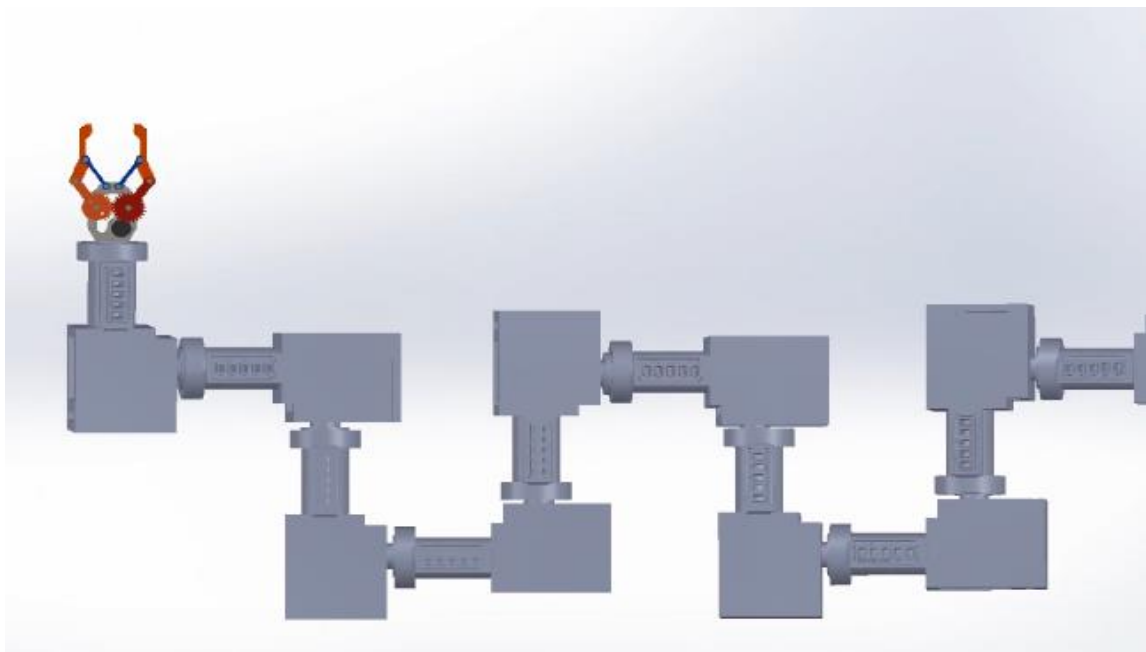


Fig 3.5.3 Degrees of freedom formed out of arrangement of modules

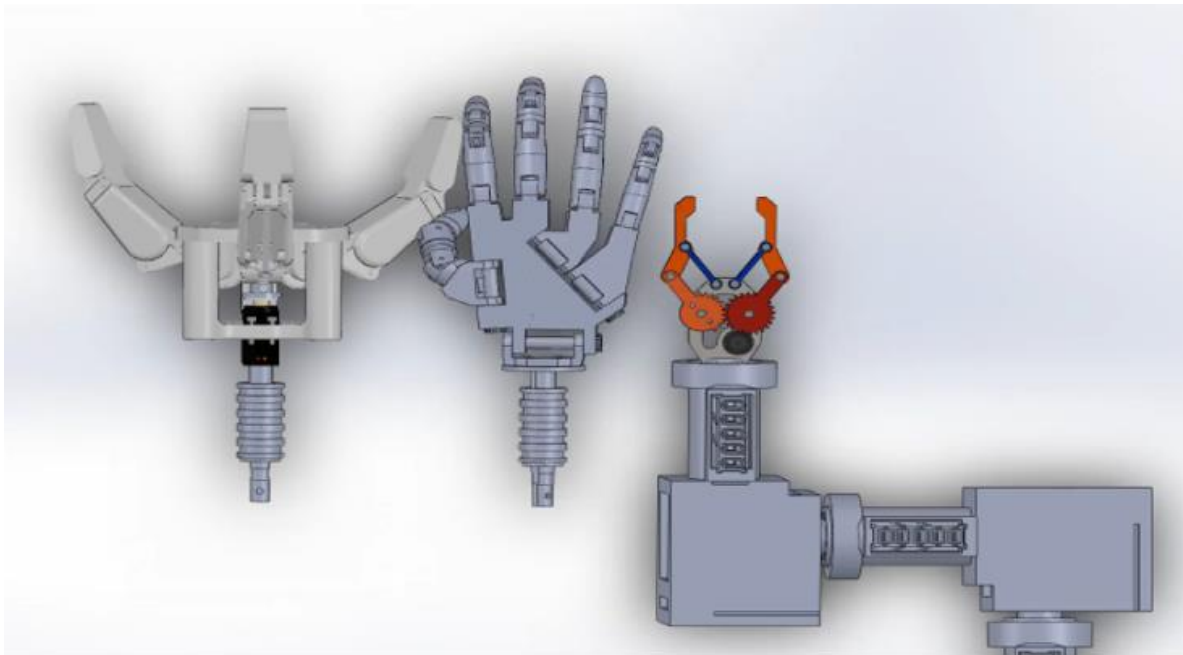


Fig 3.5.4 Different arm configurations for the tip

Designing of the robotic arm parts is done in Solid works software with at-most precision and care. Each of the module is designed with two sets of brush holders which can be assembled and detached with an ease. The brush holders provide housing for the carbon brushes and protects them from sliding and to have minimal area of contact.

The slip ring part consists 5 individual segments which are stacked one over the other and each part server their distinct purpose. Each of the slip ring segment can be detached and attached easily which gives an ease in arrangement and handling.

Modularity nature of slip ring segment makes the replacement process easy in case of damage to it. The outer frame also provides housing for stepper motor and ball bearing which facilitate frictionless rotation in the joints in between adjacent modules.

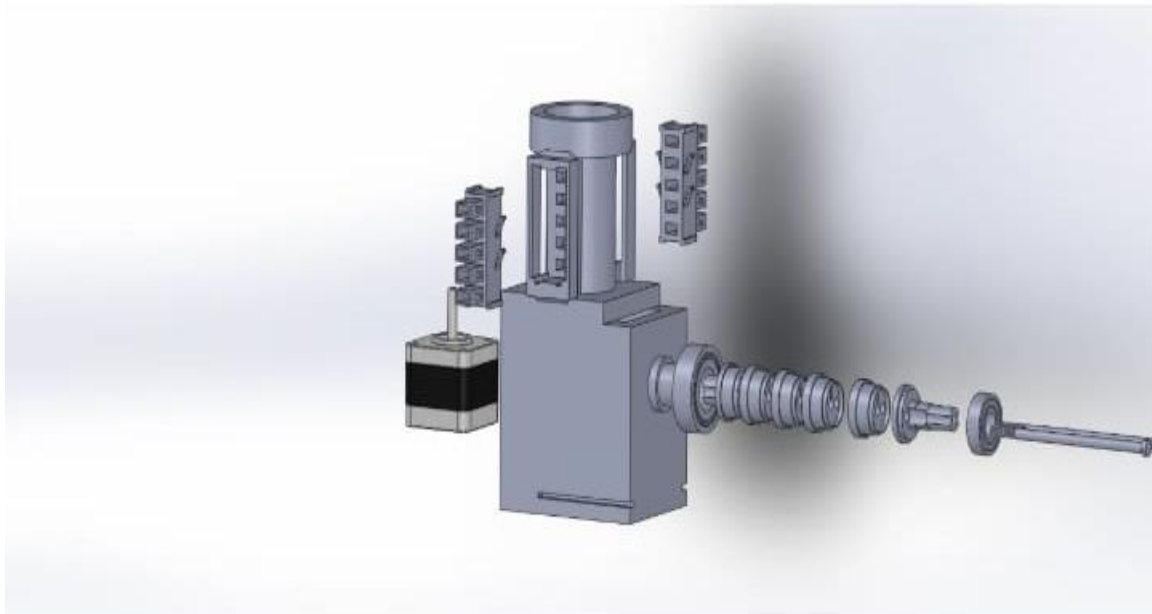


Fig 3.5.5 Interior design of each module (consisting of slip rings and carbon brushes)

3.7 Outer frame of modules (3D printed)

Rapid prototyping, also known as 3D prototyping, is not only limited to 3D printing. It actually refers to a process of iterating through a design by quickly manufacturing physical prototypes and making small changes at each step to improve it.

3D printing gives you more freedom than any other manufacturing technique. As your object is created from a digital 3D model, using a 3D modelling software, we can easily create all the designs we want, from the simplest to the most intricate ones. It is possible to create any project from scratch, get a good visualization, and to adjust your 3D model whenever you want to. 3D printing is really convenient as we don't have to create a whole mould for each iteration of the project. We are free to make different iterations, so the prototype can perfectly match our expectations for the final product.

With 3D printing, we can do as many iterations as you need. Moreover, the process itself is quite faster. Instead of the creation of the mould and waiting for the whole injection moulding process, additive manufacturing skips these steps. This manufacturing technique is perfect for rapid prototyping and can help us accelerate the whole product development.

A prototype is a way to get a perfect overview of the project to spot the last problems. It is certain to create a functional prototype. That is why, in case of having a complex project or an object with an elaborated design, be sure to check that the material and 3D printing technique is precise enough to help with the prototyping process. However, one should obviously have to keep in mind the mechanical properties of the final material that will be in use.

The material to be used in 3D printer filament is to be wisely selected by considering the facts like tensile strength, cost per module, stress applied on the material or weight carried by it. By keeping all the designing criteria and overall structure of the robotic arm in mind it is decided to make the 3D printed frame with PLA (Polylactic Acid) material, which not only posses mechanical strength but is also a bio-degradable material.

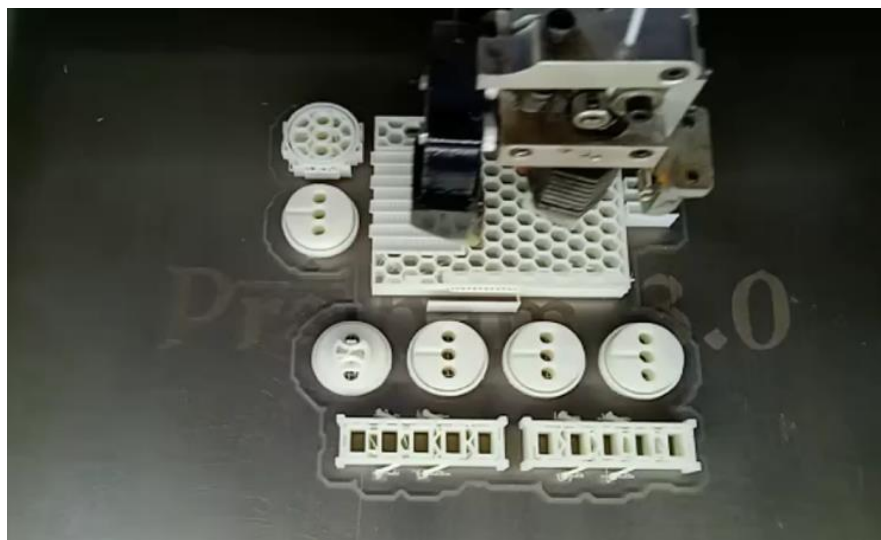


Fig 3.5.6 3D Printing of Part 1

Polylactic Acid, commonly known as PLA, is one of the most popular materials used in desktop 3D printing. It is the default filament of choice for most extrusion-based 3D printers because it can be printed at a low temperature. PLA is a great first material to use as you are learning about 3D printing because it is easy to print, very inexpensive, and creates parts that can be used for a wide variety of applications.

It is also one of the most environmentally friendly filaments on the market today. Derived from crops such as corn and sugarcane, PLA is renewable and most importantly biodegradable. This thermoplastic polymer is produced by fermenting a carbohydrate source such as corn starch. In addition, this material is considered a non-Newtonian pseudoplastic fluid. This means that its viscosity (flow resistance) will change depending on the stress to which it is subjected. Specifically, PLA is a fine cut material, which means that the viscosity decreases as you apply stress.

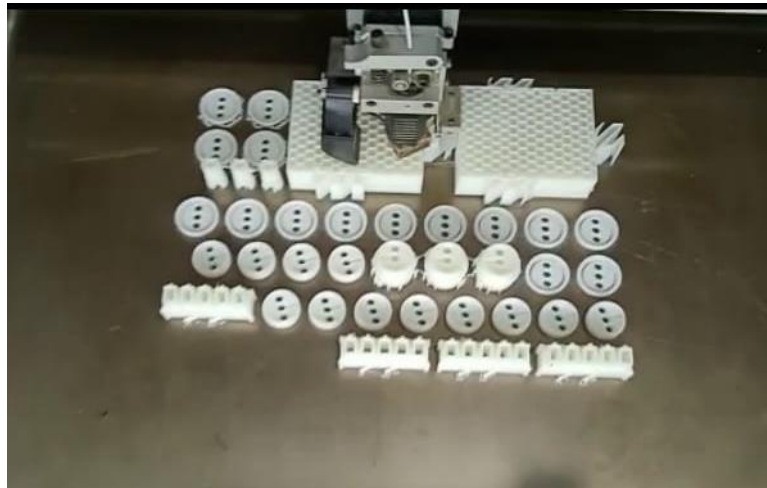


Fig 3.5.7 3D Printing of Part 2 & Part 3

PLA filament has gained wide acceptance within additive manufacturing partly because it is made from renewable products and also because of its mechanical properties. This material, considered a semi-crystalline polymer, has a melting temperature of 180°C, lower than ABS filament, which starts melting between 200°C and 260°C. This means that when printing with PLA, the use of a heated printing bed is not necessary, and the closed chamber is not a necessity either. The only drawback is that the PLA filament has a higher viscosity which can clog the print head if you are not careful.

This filament does not have the same mechanical properties as the ABS filament; the second is much more resistant and flexible. However, PLA has greater resistance to heat, so it is often used in the food industry. Even so, it is often recommended to work with it if the project does not have major mechanical complexities, as it is much simpler.

PLA does not require complex post-processing. It can be sanded or treated with acetone if required and the supports are usually very easily to remove. There may be some problems

with the first layer of this material, therefore it is recommended to add adhesive tape onto the printing tray to facilitate its detachment once the piece is finished.



Fig 3.5.8 3D Printing of Part 2 & Part 3



Fig 3.5.9 3D Printing of Part 4



Fig 3.5.10 3D Printing of Part 5



Fig 3.5.11 3D Printing of Part 6

CHAPTER 4

INCORPORATION OF STEPPER MOTOR

4.1 Introduction

To initialise the rotating motion of the arm stepper motor is installed to the system. A stepper motor, is a brushless DC electric motor that divides a full rotation into a number of equal steps. The motor's position can be commanded to move and hold at one of these steps without any position sensor , as long as the motor is correctly sized to the application in respect to torque and speed .

Stepper motor is known for the property of converting a train of input pulses into a precisely defined increment in the shaft's rotational position. Each pulse rotates the shaft through a fixed angle. Stepper motors are controlled by a driver, which sends the pulses into the motor causing it to turn.

Stepper motor suitable for robotic applications because its movement comprises of discrete steps. It moves by a discrete angle called the step angle in each step. A versatile drive system will allow accurate control of the movement of a stepper motor so that it can be used in robotic applications. The Stepper motor is best used in robots where only a certain degree of movement is required. One main problem with ordinary DC Motor is that these keep on moving round and round for infinity amount of time unless you stop feeding them with electricity.

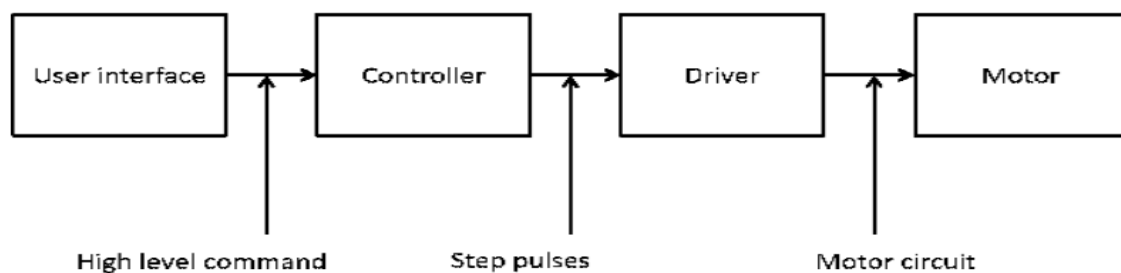


Fig.4.1.1 Block Diagram of Stepper Motor Control

The motor used in this project for the movement of robotic arm is NEMA 17 stepper motor. The stepper motors move in precisely repeatable steps, hence they are the motors of choice for the machines requiring precise position control. The Nema17 4.2 kg-cm Stepper motor can provide 4.2 kg-cm of torque at 1.7A current per phase. The motor's position can be commanded to move or hold in one position with the help of Stepper Motor Drivers.

The Nema17 4.2 kg-cm Stepper motor provides excellent response to starting, stopping, and reversing pulses from the stepper motor driver. They are very useful in various applications, especially which demands low speed with high precision. It is a brushless DC motor, so the life of this motor is dependent upon the life of the bearings. The position control is achieved by a simple Open Loop control. The motor's shaft has been machined for good grip with a pulley, drive gear, etc. and especially avoiding stall or slip.

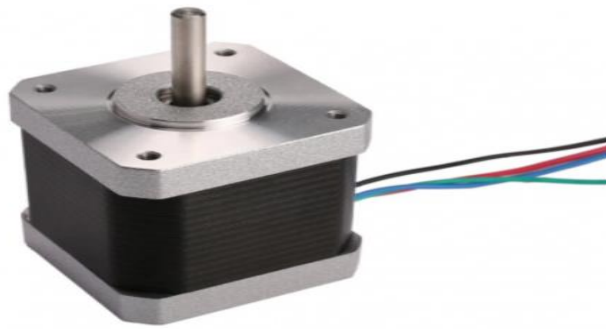


Fig 4.1.2 Stepper Motor

Specifications of the motor

1. Step Angle: 1.8°.
2. Current: 1.7 A /Phase.
3. Holding Torque: 4.2 kg-cm.
4. Voltage: 12~24VDC
5. No of Leads: 4.
6. Weight: 288 grams.

Features

1. The input pulse decides the rotation angle of the motor.
2. High accuracy of around 3 to 5% a step.
3. It provides good starting, stopping, and reversing.
4. Control of this motor is less costly because of the exclusion of complex control circuitry.
5. The speed is proportional to the frequency of the input pulses.

4.2 Stepper Motor Drive

A Stepper Motor Driver is the driver circuit that enables the stepper motor to function the way it does. For example, stepper motors require sufficient and controlled energy for phases in a precise sequence. Due to this, stepper motors are considered more advanced than the typical DC motor. The driver circuit that controls how the stepper motor operates.

Stepper drives work by sending current through various phases in pulses to the stepper motor. A simple way to choose a stepper drive is to look for four things voltage, current, micro stepping, and maximum step pulse rate. It should be ensured that the drive can handle a wide range of current so that the system can be tested at different voltage levels to fit our application.

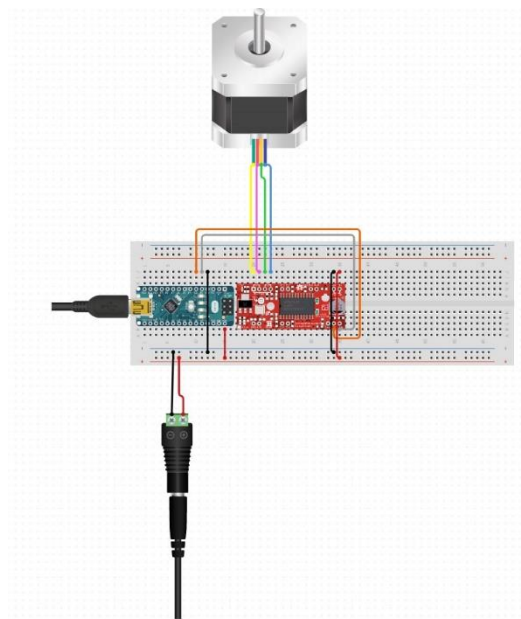


Fig.4.2.1 Block diagram of stepper motor driver circuit

Stepper motor drivers rely on input from a separate source in order to function and create an output.

The following four signals control the motor:

1. CLOCK: The clock signal gives the stepping command.
2. RESET: Puts the final level signals in a defined start position.
3. DIRECTION: Determines the sense of rotation of the motor axis.
4. HALF/ FULL: Decides whether to operate in full or in half-step.

In this project we use motor driver LN298 to drive the motor. As L298N module has two H-Bridges, each H-Bridge will drive one of the electromagnetic coils of a stepper motor. By energizing these electromagnetic coils in a specific sequence, the shaft of a stepper can be moved forward or backward precisely in small steps.

This L298N Motor Driver Module is a high-power motor driver module for driving DC and Stepper Motors. This module consists of an L298 motor driver IC and a 78M05 regulator. L298N Module can control up to 4 DC motors, or 2 DC motors with directional and speed control.

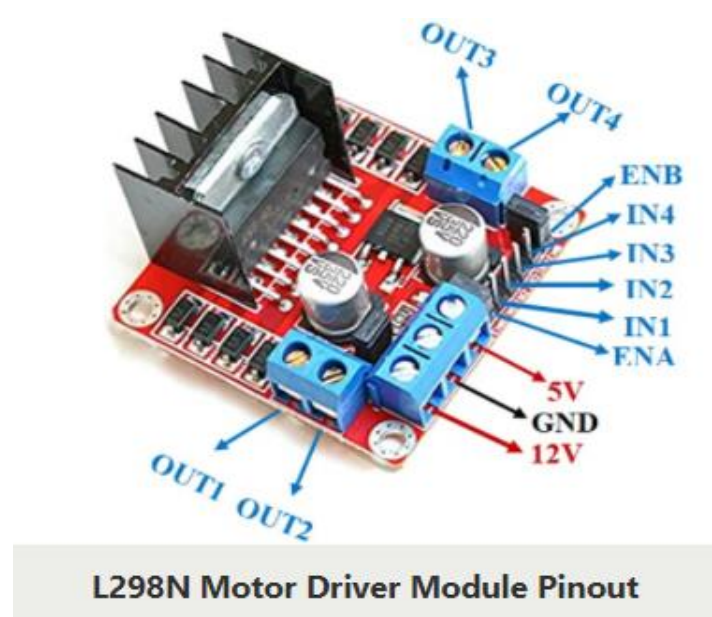


Fig 4.2.2 L298N Motor Driver

L298N Module Pinout Configuration

1. **1.IN1 & IN2:** Motor A input pins.

Used to control the spinning direction of Motor A

2. **IN3 & IN4:** Motor B input pins.

Used to control the spinning direction of Motor B

3. **ENA:** Enables PWM signal for Motor A

4. **ENB:** Enables PWM signal for Motor B

5. **OUT1 & OUT2:** Output pins of Motor A

6. **OUT3 & OUT4:** Output pins of Motor B

7. **12V:** 12V input from DC power Source

8. **5V:** Supplies power for the switching logic circuitry inside L298N IC

9. **GND:** Ground pin

Features & Specifications

- Driver Model: L298N 2A
- Driver Chip: Double H Bridge L298N
- Motor Supply Voltage (Maximum): 46V
- Motor Supply Current (Maximum): 2A
- Logic Voltage: 5V
- Driver Voltage: 5-35V
- Driver Current: 2A
- Logical Current: 0-36mA
- Maximum Power (W): 25W
- Current Sense for each motor
- Heatsink for better performance
- Power-On LED indicator.

The L298N Motor Driver module consists of an L298 Motor Driver IC, 78M05 Voltage Regulator, resistors, capacitor, Power LED, 5V jumper in an integrated circuit. 78M05 Voltage regulator will be enabled only when the jumper is placed. When the power supply is less than or equal to 12V, then the internal circuitry will be powered by the voltage regulator and the 5V pin can be used as an output pin to power the microcontroller. The jumper should not be placed

when the power supply is greater than 12V and separate 5V should be given through 5V terminal to power the internal circuitry.

ENA & ENB pins are speed control pins for Motor A and Motor B while IN1& IN2 and IN3 & IN4 are direction control pins for Motor A and Motor B.

CHAPTER 5

SERIAL INTERFACING

5.1 INTRODUCTION

One of the easiest and inexpensive way to control stepper motors is to interface L298N Motor Driver with Arduino. It can control both speed and spinning direction of any Bipolar stepper motor like NEMA 17.

As L298N module has two H-Bridges, each H-Bridge will drive one of the electromagnetic coils of a stepper motor. By energizing these electromagnetic coils in a specific sequence, the shaft of a stepper can be moved forward or backward precisely in small steps. However, the speed of a motor is determined by the how frequently these coils are energized.

Before we start hooking the motor up with the module, we have to determine the A+, A-, B+ and B- wires on the motor. For our motor these are red, green, blue and yellow.

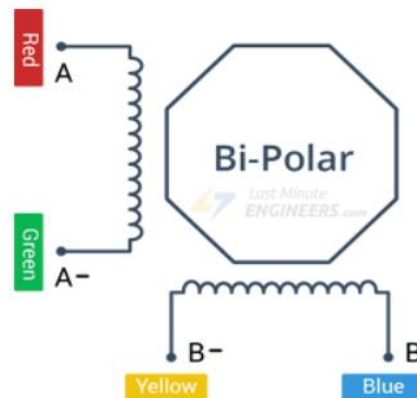


Fig 5.1.1 Internal Connection of Stepper Motor

Start by connecting external 12V power supply to the VCC terminal. And keep the 5V-EN jumper in place. We need to keep both the ENA & ENB jumpers in place so the motor is always enabled.

Now, connect the input pins (IN1, IN2, IN3 and IN4) of the L298N module to four Arduino digital output pins (8, 9, 10 and 11). Finally, connect the A+, A-, B+ and B- wires from the stepper motor to the module

5.2 Arduino Uno

The Arduino Uno is a microcontroller board based on the AT mega 328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with an AC-to-DC adapter or battery to get started. The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter.

"Uno" means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduino boards, and the reference model for the Arduino platform; for a comparison with previous versions, see the index of Arduino boards.

Power Supply

The Arduino Uno can be powered via the USB connection or with an external power supply. The power source is selected automatically. External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery.

VIN: The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.

- **5V:** This pin outputs a regulated 5V from the regulator on the board. The board can be supplied with power either from the DC power jack (7 - 12V), the USB connector (5V), or the VIN pin of the board (7-12V). Supplying voltage via the 5V or 3.3V pins bypasses the regulator, and can damage your board. We don't advise it.
- **3V-** A 3volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- **GND-** Ground pins.

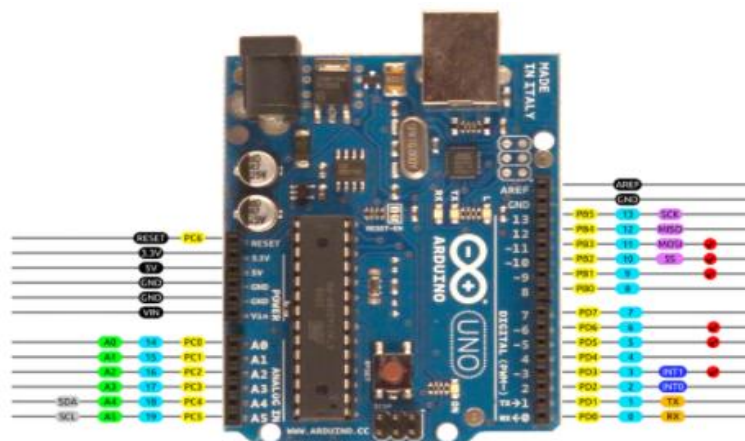


Fig 5.2.1 Arduino circuit

Pin Configuration

- **Serial:** 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip.
- **External Interrupts:** 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the attach Interrupt () function for details.
- **PWM:** 3, 5, 6, 9, 10, and 11. Provide 8-bit PWM output with the analog write () function.
- **SPI:** 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK). These pins support SPI communication using the SPI library.
- **LED:** 13. There is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off. The Uno has 6 analog inputs, labelled A0 through A5, each of which provide 10 bits of resolution.
- **TWI:** A4 or SDA pin and A5 or SCL pin. Support TWI communication using the Wire library. There are a couple of other pins on the board:
- **AREF:** Reference voltage for the analog inputs. Used with default, ().
- **Reset:** Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

5.3 Arduino Nano

Arduino Nano is a small, complete, and breadboard-friendly development board based on the ATmega328 (Arduino Nano 3.0) or ATmega168 (Arduino Nano 2.x). It has more or less the same functionality of the Arduino UNO, but in a different package. It lacks only a DC power jack, and works with a Mini-B USB cable instead of a standard one. The connectivity is the same as the Arduino UNO board.

NANO has the same functionality and the same features except the number of pins and size than other variant in its family. Due to compact size, the only limitation of this board is it doesn't have any input power jack. So, you can't supply power from any external power source like a battery. Apart from this, more or less this board is quite similar to any Arduino board.

This development board provides small plate with the same powerful microcontroller as on the Arduino Uno. So, it is compatible with all Arduino ecosystem and ideal for fast prototyping and easy integration, It can be used with other sensors to create different montages.

Nano is Ideal for small projects and has the advantage that it can be programmed via USB or even you can programme it using your Android phone. Although there is no official IDE for Android yet but makers community find the way to it.

Specifications

- It has 22 input/output pins in total.
- 14 of these pins are digital pins.
- Arduino Nano has 8 analogue pins.
- It has 6 PWM pins among the digital pins.
- It comes with 16MHz clock speed via a crystal oscillator
- Its operating voltage varies from 5V to 12V.
- It also supports different ways of communication, which are:
 - a. Serial Protocol.
 - b. I2C Protocol.
 - c. SPI Protocol.
- It also has a mini-USB Pin which is used to upload code

Memory in Arduino Nano

It has below memories embedded in it which are used for different purposes and are as follows:

- Flash memory of Arduino Nano is 32Kb.
- It has preinstalled bootloader on it, which takes a flash memory of 2kb.
- SRAM memory of this Microcontroller board is 8kb.
- It has an EEPROM memory of 1kb.
- It has a Reset button on it.

Pin Configuration

- **Power Pin (Vin, 3.3V, 5V, GND):**
 - a. **Vin** - input voltage of the board, and it is used when an external power source is used from 7V to 12V.
 - b. **5V** - regulated power supply voltage of the nano board and it is used to give the supply to the board as well as components.
 - c. **3V** - minimum voltage which is generated from the voltage regulator on the board.
 - d. **GND** - ground pin of the board
- **RST Pin (Reset):** This pin is used to reset the microcontroller
- **Analog Pins (A0-A7):** These pins are used to calculate the analog voltage of the board within the range of 0V to 5V
- **I/O Pins (Digital Pins from D0 – D13):** These pins are used as an i/p otherwise o/p pins. 0V & 5V
- **Serial Pins (Tx, Rx):** These pins are used to transmit & receive TTL serial data.
- **External Interrupts (2, 3):** These pins are used to activate an interrupt.
- **PWM (3, 5, 6, 9, 11):** These pins are used to provide 8-bit of PWM output.
- **SPI (10, 11, 12, & 13):** These pins are used for supporting SPI communication.
- **Inbuilt LED (13):** This pin is used to activate the LED.
- **IIC (A4, A5):** These pins are used for supporting TWI communication.
- **AREF:** This pin is used to give reference voltage to the input voltage.

Serial communication interface is also known as Serial Peripheral Interface. It provisions full duplex communication mode. It means data can be received and sent simultaneously. Master can send data to slaves and slaves can send data to master at same time. To transmit and receive data, a clock signal is required which means it is a synchronous communication channel.

- SCK = Serial Clock = This line is used for clock pulses to synchronize data transmission from Master devices
- MOSI = Master Out Slave In = This line is used for sending data from Master device to Slave devices.
- MISO = Master In Slave Out = This line is used for sending data from Slave devices to Master device.
- SS = Slave Select = This pin is used by Master device to enable and disable specific devices to communicate with only some number of devices out of all.
 - Pin11 = MOSI
 - Pin12 = MISO
 - Pin13 = SCK
 - Pin10 = SS

To establish communication between two Arduino using SPI, there is a pre-built library of Arduino used for this purpose

- **<SPI.h>**: It includes the library in the program to use the function of SPI communication.
- **SPI.begin()**: It initializes the SPI bus and sets the SCK and MOSI pin to LOW and SS pin to HIGH.
- **SPI.transfer(Val)**: This is used for simultaneous communication between Master and Slave devices.
- **SPI.attachInterrupt(handler)**: It is used to receive data from a master device in a slave device and is called in a slave device.
- **SPI.setClockDivider(divider)**: It set the SPI clock divider in accordance with the system clock.

In-order to perform SPI communication between two Arduino boards we will configure one Arduino as a master device and another Arduino board as a slave device. A master device will transmit 1 if the user pressed the push button which is connected with the D2 of Arduino. SPI Slave Arduino receives this data and based on received data make a decision to turn-on or turn-off LED which is connected with the D4 pin of Arduino.

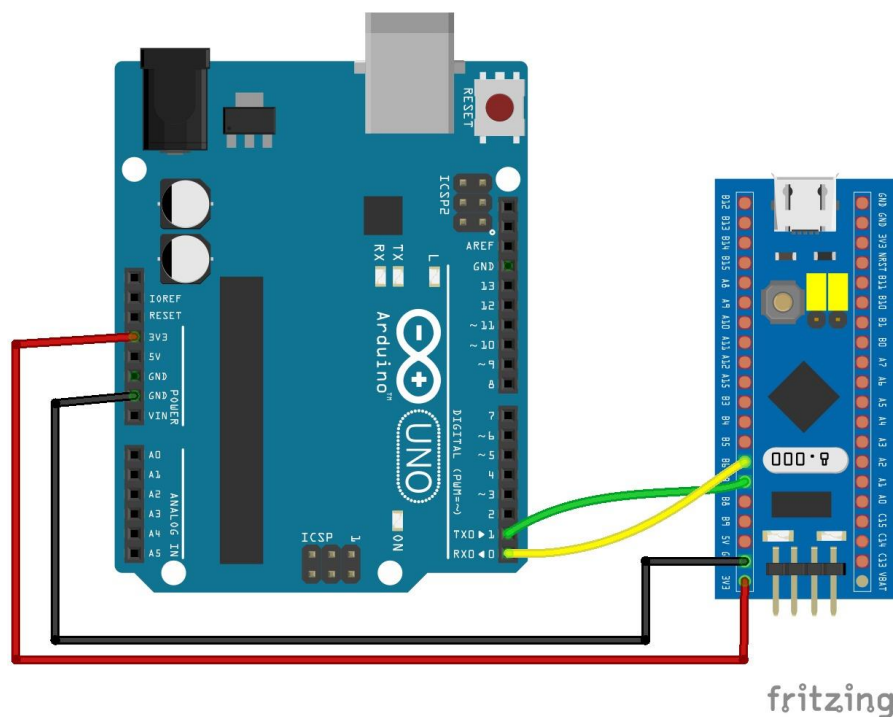


FIG 5.3.1 Serial Connection Between Arduino Nano And Uno

CHAPTER 6

AUGMENTED REALITY

6.1 Introduction

Augmented reality (AR) is an enhanced version of the real physical world that is achieved through the use of digital visual elements, sound, or other sensory stimuli delivered via technology. One of augmented reality's primary goals is to highlight specific features of the physical world, increase understanding of those features, and derive smart and accessible insight that can be applied to real-world applications. Augmented reality is an interactive experience of a real-world environment where the objects that reside in the real world are enhanced by computer-generated perceptual information, sometimes across multiple sensory modalities, including visual, auditory, haptic, etc.

Today augmented reality has not only spread its wings into gaming and entertainment but also into our day-to-day life. This technology has shown its capabilities and showcased its future scopes to all of us, over the recent course of time. Currently, Augmented reality has a vast number of application areas in various sectors and categories. With each sector getting explored with the fullest scope of implementing AR technology from retail businesses to health sectors.

6.2 Scope for AR

Robotics

In robotics, AR acts as a new medium for interaction and information exchange with autonomous systems increasing the efficiency of the Human-Robot Interaction (HRI).

Space research

Augmented reality headsets have become a staple on the International Space Station, used for everything from controlling robots to carrying out maintenance and even exercising in space.

Instead of using a massive instruction and assembly manual, engineers can use AR to see the instructions, what parts they need and even the next steps projected right onto the surface where they work.

Surgical Training

Hands-on experience is best, but the last thing you want is a new surgeon carrying out a delicate and dangerous operation on their first day. AR can give physicians the experience they need so they're ready to save lives when the need arises.

Property Tours and Project Inspections

The COVID-19 pandemic has made face-to-face interactions a challenge. AR gives people the ability to tour properties without ever leaving the comfort of home.

Training and Education

Dynamic, AR-based instructions let people perform new tasks more easily and quickly than traditional training methods (like instruction manuals).

Open CV

OpenCV is a cross-platform library using which we can develop real-time computer vision applications. It mainly focuses on image processing, video capture and analysis including features like face detection and object detection. Here Open CV is used to detect palm landmarks, which are used to vectorize and calculate the joints if the gripper does grab and releasing action. From Open CV command is being send to Arduino NANO which is used for the coordinated operation of the system. In addition to these a servo motor is connected in-order to facilitate position control.

CHAPTER 7

Assembly of slip ring



Fig 7.1.1 slip ring

Copper strip and 3D printed parts are assembled using a wire and screw to hold them. The copper rings used to conduct through the wires. Slip ring is located at the centre of the part. It is held by the stepper motor.

Connections of Arduino nano and stepper motor driver in perforated board

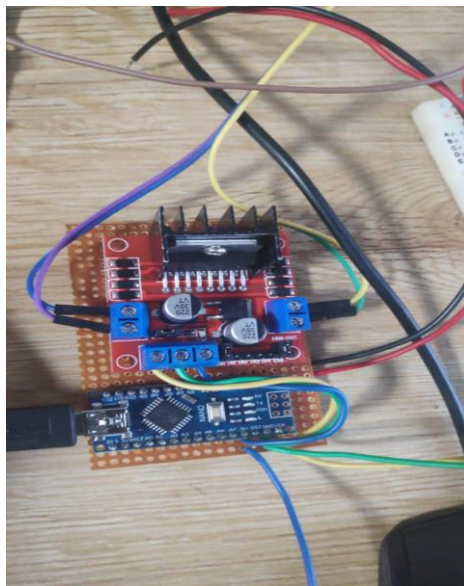


Fig 7.1.2 Arduino nano and stepper motor in perforated board

Arduino nano and motor drive are soldered to the perforated board. There are four pins 12V, ground, RX and TX. The perforated circuit board is housed in the main body of the 3D printed part. The code is uploaded to the Arduino nano through USB.

Assembly of carbon brushes

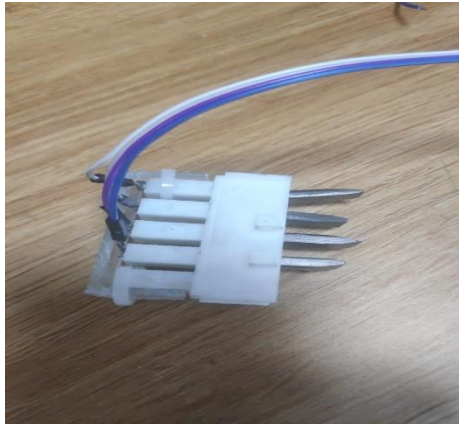


Fig 7.1.3 Carbon brushes

There are four carbon brushes which need to be grinded down to a suitable size of the 3D printed brush holder. Using acrylic sheets, a cap for each 3D printed brush holder is made. The carbon brush holder is housed in the two sides of the main 3D printed part.

Assembly of a part



Fig 7.1.4 Assembly of part of robotic arm

The stepper motor is placed inside the 3D part. 3D printed Slip ring is attached to the shaft of the stepper motor. Carbon brushes holder are attached to each side of the main part.

Computer setup for AR vision

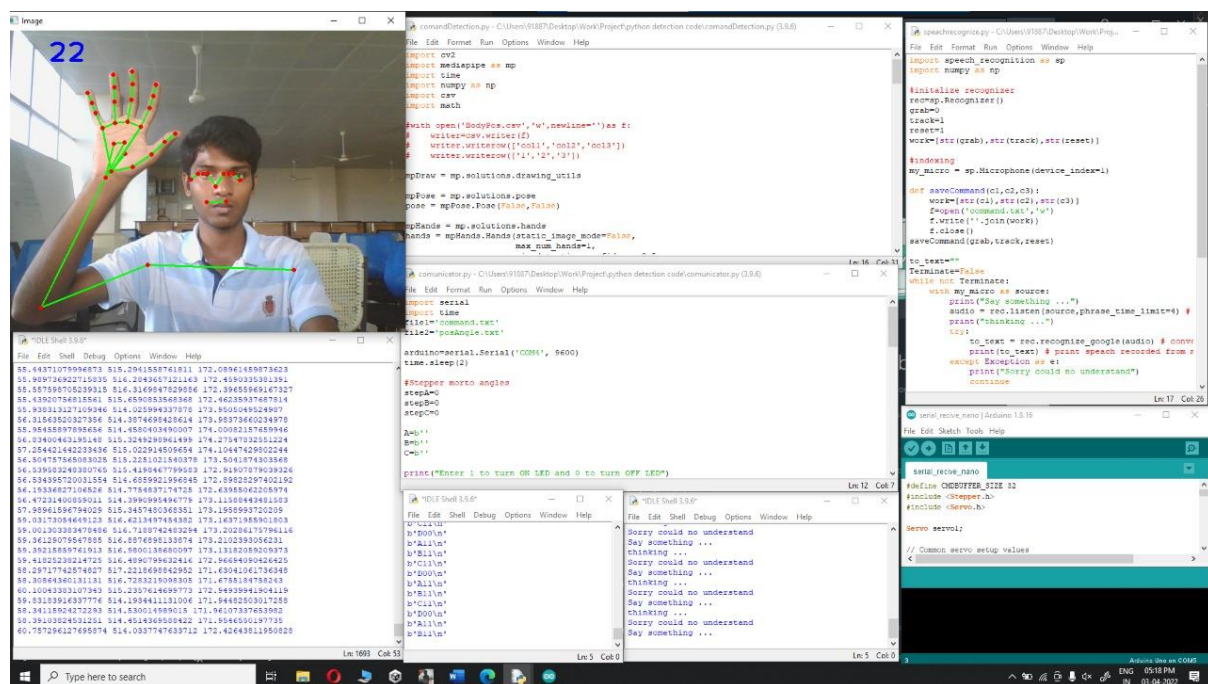


Fig 7.1.5 AR vision computer setup

Commanddetection.py code is run first which is used to detect the person's body gestures through camera. Speechrecognition.py code is run to initialize the position and get further voice commands from the user. Serial communication between computer and Arduino is run using the code in communication.py.

Stepper motor testing

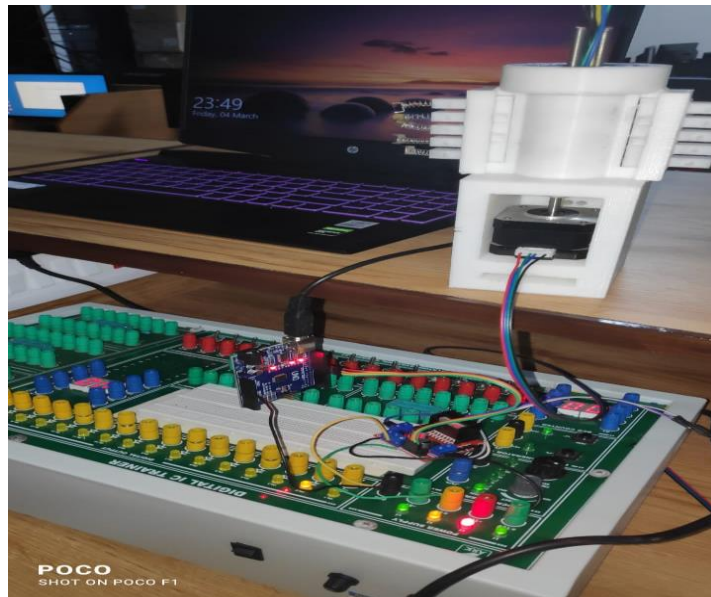


Fig 7.1.6 Testing of stepper motor

The code for Arduino is uploaded and computer is setup with the visual inputs. Body gesture of the person is taken as coordinates to move the stepper motor.

Ex: When a person moves left, the motor will rotate clockwise and when the person moves right, the motor will rotate anticlockwise.

Servomotor and serial communication testing

The code for Arduino is uploaded and computer is setup with the visual inputs. Hand gestures of the person is taken as input for servomotor.

Ex: When a person shows his palm, the servomotor opens its claw and when the person shows his fist the servomotor will close its claw.

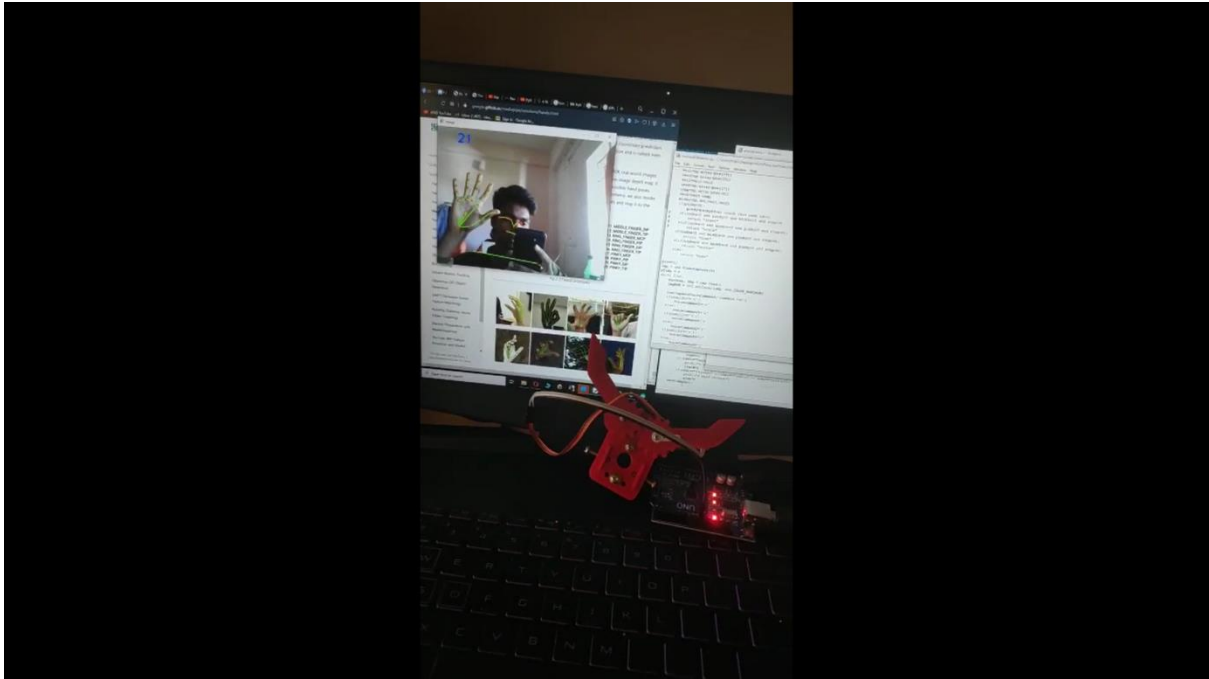


Fig 7.1.7 Controlling of gripper movement using AR technology (Opening)

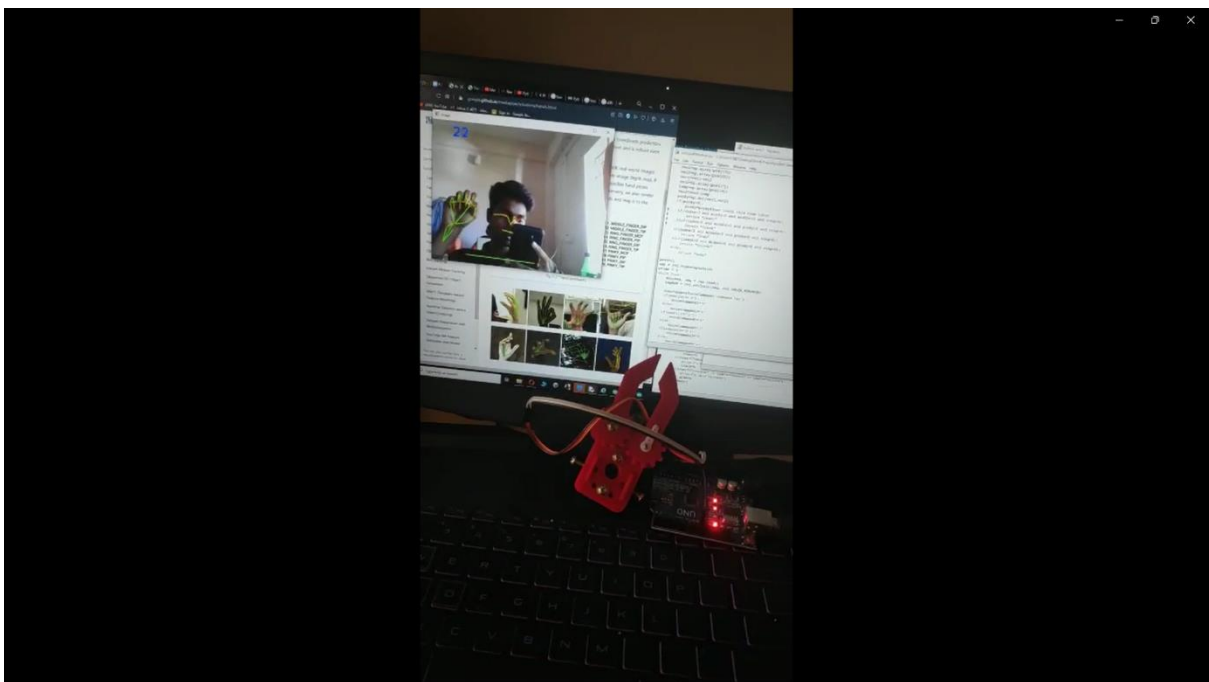


Fig 7.1.8 Controlling of gripper using AR technology (Closing)

Assembly of robotic arm



Fig 7.1.9 Assembly of robotic arm

The three parts are assembled which includes two stepper motor and one servomotor.

CHAPTER 8

CONCLUSION

This project discusses about the challenges faced with conventional robots. Some of the problems faced by conventional robots are their static nature, hardships in rebuilding and reprogramming and their applications are of narrow range. In this regard we came up with solutions of designing and development of robotic arm with reassembling capabilities which is recent trend in advancement of robots. Reconfigurable robots have the ability to be arranged in a correct range of configuration according to the user's desires.

The main objective of this project is to design and development of individual module in which current condition is done by slip rings arrangement. The modular fabricated in a 3D printer and later the module are arranged to a complete arm structure. This modularity and reconfigurability nature of the system gives base in replacement of individual modules and also can be utilised in wide range of applications.

The arm is given a gripper to perform simple pick and place operation which is controlled by AR technology. Advantages of this modular robot is that it is capable of facilitating change in the configurations according to environmental conditions. It can be Re-configured to obtain flexibility in a single robot to perform multiple tasks. Functional advantages are its morphologically adaptive nature and its potentially robust system.

Further Proceeding:

Bluetooth or WIFI module - using of Bluetooth or Wi-Fi module instead of current serial communication can reduce the number of sliprings used

contactless slipring - contactless slipring is designed for deterministic data transmission in real time and continuous operation, this can reduce the size and help efficient transfer of data

easy to handle design - current design has a multistep assemble and deassemble procedures. this can be reduced in future designs

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